

μ 's Paper 3

Markscheme

Instructions to examiners

- **M** marks are awarded for attempting to use a correct method.
- **A** marks are awarded for accuracy or for a correct answer.
- **R** marks are awarded for clear reasoning.
- **AG** means the answer is given in the question.
- Follow-through marks may be awarded where appropriate.
- Equivalent forms of answers should be accepted unless a specific form is required.

May 2026

1.

(a) (i)

$$PA = \sqrt{(x+a)^2 + y^2}$$

or

$$PB = \sqrt{(x-a)^2 + y^2}$$

A1

$$\sqrt{(x+a)^2 + y^2} \sqrt{(x-a)^2 + y^2} = b^2$$

A1

$$\boxed{\sqrt{(x+a)^2 + y^2} \sqrt{(x-a)^2 + y^2} = b^2}$$

[2 marks]

(ii) Squaring both sides,

$$\left(\sqrt{(x+a)^2 + y^2} \sqrt{(x-a)^2 + y^2} \right)^2 = (b^2)^2$$

$$((x+a)^2 + y^2)((x-a)^2 + y^2) = b^4$$

A1

$$\boxed{((x+a)^2 + y^2)((x-a)^2 + y^2) = b^4}$$

[1 mark]

(b)

$$(x+a)^2 + y^2 = x^2 + y^2 + a^2 + 2ax$$

$$(x-a)^2 + y^2 = x^2 + y^2 + a^2 - 2ax$$

$$((x+a)^2 + y^2)((x-a)^2 + y^2) = (x^2 + y^2 + a^2 + 2ax)(x^2 + y^2 + a^2 - 2ax)$$

M1

continued...

Let

$$U = x^2 + y^2 + a^2.$$

Then

$$(x^2 + y^2 + a^2 + 2ax)(x^2 + y^2 + a^2 - 2ax) = (U + 2ax)(U - 2ax)$$

$$= U^2 - (2ax)^2$$

A1

$$= (x^2 + y^2 + a^2)^2 - 4a^2x^2$$

A1

Hence, using part (a)(ii),

$$(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4$$

A1

$$\boxed{(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4}$$

[4 marks]

Note: Allow any sensible reasoning leading to the required answer.

(c) The equation from part (b) is

$$(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4.$$

Replacing y by $-y$ leaves the equation unchanged since y appears only as y^2 .

R1

Therefore, the curve is symmetric about the x -axis.

Replacing x by $-x$ leaves the equation unchanged since x appears only as x^2 .

R1

Therefore, the curve is symmetric about the y -axis.

[2 marks]

continued...

(d) (i) Set $y = 0$.

$$((x + a)^2)((x - a)^2) = b^4$$

M1

$$(x + a)^2(x - a)^2 = b^4$$

$$((x + a)(x - a))^2 = b^4$$

$$(x^2 - a^2)^2 = b^4$$

Taking square roots,

$$\boxed{|x^2 - a^2| = b^2}$$

A1

Note: Do **not** award **A1** if the absolute value is missing.

continued...

(ii) From

$$|x^2 - a^2| = b^2,$$

we have

$$\begin{array}{ll} x^2 - a^2 = b^2 & \text{or} \quad x^2 - a^2 = -b^2 \\ \therefore x^2 = a^2 + b^2 & \text{or} \quad x^2 = a^2 - b^2 \end{array}$$

A1

Therefore,

$$x = \pm\sqrt{a^2 + b^2},$$

and

$$x = \pm\sqrt{a^2 - b^2}$$

So the possible x -intercepts are

$$\boxed{(\pm\sqrt{a^2 + b^2}, 0)}$$

and, if $b \leq a$,

$$\boxed{(\pm\sqrt{a^2 - b^2}, 0)}.$$

A1

Note: Conditions on a and b are not necessary for the two **A** marks above.

(iii) Set $x = 0$.

$$(y^2 + a^2)^2 = b^4$$

M1

Since $y^2 + a^2 > 0$,

$$y^2 + a^2 = b^2$$

$$y^2 = b^2 - a^2.$$

For intersections with the y -axis to exist,

$$b^2 - a^2 \geq 0.$$

Hence,

$$b \geq a.$$

A1

$$\boxed{b \geq a}$$

[6 marks]

(e) When $b = a$, the equation from part (b) becomes

$$(x^2 + y^2 + a^2)^2 - 4a^2x^2 = a^4$$

M1

Expanding,

$$(x^2 + y^2)^2 + 2a^2(x^2 + y^2) + a^4 - 4a^2x^2 = a^4$$

A1

Cancelling a^4 ,

$$(x^2 + y^2)^2 + 2a^2(x^2 + y^2) - 4a^2x^2 = 0$$

$$(x^2 + y^2)^2 + 2a^2x^2 + 2a^2y^2 - 4a^2x^2 = 0$$

$$(x^2 + y^2)^2 - 2a^2x^2 + 2a^2y^2 = 0$$

A1

Therefore,

$$(x^2 + y^2)^2 = 2a^2x^2 - 2a^2y^2$$

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$$

A1

$$\boxed{(x^2 + y^2)^2 = 2a^2(x^2 - y^2)}$$

[4 marks]

(f) Using

$$x = r \cos \theta, \quad y = r \sin \theta,$$

$$x^2 + y^2 = r^2$$

A1

continued...

Also,

$$x^2 - y^2 = r^2 \cos^2 \theta - r^2 \sin^2 \theta$$

$$= r^2 (\cos^2 \theta - \sin^2 \theta)$$

$$= r^2 \cos(2\theta)$$

A1

Substituting into

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2),$$

$$r^4 = 2a^2 r^2 \cos(2\theta)$$

For $r \neq 0$,

$$r^2 = 2a^2 \cos(2\theta)$$

A1

$$\boxed{r^2 = 2a^2 \cos(2\theta)}$$

[3 marks]

Note: The origin is included in the curve, so division by r^2 does not lose the required polar representation.

(g) METHOD 1

For the right-hand loop,

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}.$$

Using

$$y = r \sin \theta,$$

$$y^2 = r^2 \sin^2 \theta$$

M1

continued...

Since

$$r^2 = 2a^2 \cos(2\theta),$$

$$y^2 = 2a^2 \cos(2\theta) \sin^2 \theta$$

A1

Therefore,

$$f(\theta) = 2a^2 \sin^2 \theta \cos(2\theta)$$

A1

Let

$$u = \sin^2 \theta.$$

Using

$$\cos(2\theta) = 1 - 2 \sin^2 \theta,$$

$$\cos(2\theta) = 1 - 2u$$

M1

Hence,

$$y^2 = 2a^2 u(1 - 2u)$$

$$y^2 = 2a^2(u - 2u^2)$$

A1

Now maximize

$$u - 2u^2.$$

$$\frac{d}{du}(u - 2u^2) = 1 - 4u$$

continued...

Set

$$1 - 4u = 0$$

M1

$$u = \frac{1}{4}$$

Substitute into the expression for y^2 :

$$y^2 = 2a^2 \left(\frac{1}{4} \right) \left(1 - 2 \cdot \frac{1}{4} \right)$$

$$y^2 = 2a^2 \left(\frac{1}{4} \right) \left(\frac{1}{2} \right)$$

$$y^2 = \frac{a^2}{4}.$$

Since the question asks for the maximum value of y ,

$$y_{\max} = \frac{a}{2}$$

A1

$$\boxed{y_{\max} = \frac{a}{2}}$$

[7 marks]

METHOD 2

Starting from

$$y^2 = 2a^2 \sin^2 \theta \cos(2\theta),$$

differentiate with respect to θ .

$$\frac{d}{d\theta}(y^2) = 2a^2 (2 \sin \theta \cos \theta \cos(2\theta) - 2 \sin^2 \theta \sin(2\theta))$$

M1A1

Using

$$\sin(2\theta) = 2 \sin \theta \cos \theta,$$

$$\frac{d}{d\theta}(y^2) = 4a^2 \sin \theta \cos \theta (\cos(2\theta) - 2 \sin^2 \theta)$$

A1

Set

$$\frac{d}{d\theta}(y^2) = 0.$$

M1

The non-zero interior solution satisfies

$$\cos(2\theta) - 2 \sin^2 \theta = 0.$$

So

$$\cos(2\theta) = 2 \sin^2 \theta.$$

Using

$$\cos(2\theta) = 1 - 2 \sin^2 \theta,$$

$$1 - 2 \sin^2 \theta = 2 \sin^2 \theta$$

$$\sin^2 \theta = \frac{1}{4}$$

A1

Substitute back into

$$y^2 = 2a^2 \sin^2 \theta \cos(2\theta).$$

Since

$$\sin^2 \theta = \frac{1}{4},$$

$$\cos(2\theta) = 1 - 2 \left(\frac{1}{4} \right) = \frac{1}{2}.$$

Therefore,

$$y^2 = 2a^2 \left(\frac{1}{4} \right) \left(\frac{1}{2} \right)$$

$$y^2 = \frac{a^2}{4}$$

A1

Hence,

$$y_{\max} = \frac{a}{2}$$

A1

$$\boxed{y_{\max} = \frac{a}{2}}$$

Note: Accept equivalent methods, including completing the square or using a GDC to maximize $f(\theta)$, provided sufficient working is shown.

[7 marks]

2.

(a)

$$a_1 = \sqrt{2 + a_0} = \sqrt{2}$$

A1

$$a_2 = \sqrt{2 + a_1} = \sqrt{2 + \sqrt{2}}$$

$$a_3 = \sqrt{2 + a_2} = \sqrt{2 + \sqrt{2 + \sqrt{2}}}$$

A1

$a_1 = \sqrt{2},$	$a_2 = \sqrt{2 + \sqrt{2}},$	$a_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}$
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[2 marks]

Note: Award the second **A1** only if both a_2 and a_3 are correct. Accept equivalent exact forms.
continued...

(b) (i) Using the double-angle identity,

$$\cos \theta = 2 \cos^2 \left(\frac{\theta}{2} \right) - 1$$

$$2 + 2 \cos \theta = 2 + 2 \left(2 \cos^2 \left(\frac{\theta}{2} \right) - 1 \right)$$

M1

$$2 + 2 \cos \theta = 4 \cos^2 \left(\frac{\theta}{2} \right)$$

Since $0 \leq \theta \leq \pi$,

$$0 \leq \frac{\theta}{2} \leq \frac{\pi}{2},$$

so

$$\cos \left(\frac{\theta}{2} \right) \geq 0.$$

Therefore,

$$\sqrt{2 + 2 \cos \theta} = \sqrt{4 \cos^2 \left(\frac{\theta}{2} \right)} = 2 \cos \left(\frac{\theta}{2} \right)$$

A1

$$\sqrt{2 + 2 \cos \theta} = 2 \cos \left(\frac{\theta}{2} \right)$$

[2 marks]

Note: The condition $0 \leq \theta \leq \pi$ is needed to justify taking the positive square root.

continued...

(ii) For $n = 0$,

$$a_0 = 0$$

and

$$2 \cos \left(\frac{\pi}{2^{0+1}} \right) = 2 \cos \left(\frac{\pi}{2} \right) = 0.$$

So the result is true for $n = 0$.

A1

Assume that the result is true for some integer $n = k$, where $k \geq 0$. That is,

$$a_k = 2 \cos \left(\frac{\pi}{2^{k+1}} \right).$$

M1

Then

$$a_{k+1} = \sqrt{2 + a_k}$$

M1

Using the induction hypothesis,

$$a_{k+1} = \sqrt{2 + 2 \cos \left(\frac{\pi}{2^{k+1}} \right)}.$$

A1

Using part (b)(i), with

$$\theta = \frac{\pi}{2^{k+1}},$$

$$a_{k+1} = 2 \cos \left(\frac{1}{2} \cdot \frac{\pi}{2^{k+1}} \right) = 2 \cos \left(\frac{\pi}{2^{k+2}} \right).$$

A1

Thus the result is true for $n = k + 1$.

Since the result is true for $n = 0$, and truth for $n = k$ implies truth for $n = k + 1$, the result is true for all integers $n \geq 0$.

R1

$$\boxed{a_n = 2 \cos \left(\frac{\pi}{2^{n+1}} \right)}$$

[6 marks]

Note: Award **M0** for statements such as “let $n = k$ ” unless the assumption that the statement is true for $n = k$ is made clear.

continued...

(c) (i) Using part (b)(ii),

$$\frac{a_k}{2} = \cos\left(\frac{\pi}{2^{k+1}}\right).$$

Therefore,

$$P_n = \prod_{k=1}^n \frac{a_k}{2} = \prod_{k=1}^n \cos\left(\frac{\pi}{2^{k+1}}\right)$$

A1

$$P_n = \prod_{k=1}^n \cos\left(\frac{\pi}{2^{k+1}}\right)$$

[1 mark]

continued...

(ii) METHOD 1

Start with

$$1 = \sin\left(\frac{\pi}{2}\right)$$

and use

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

repeatedly.

M1

First,

$$\sin\left(\frac{\pi}{2}\right) = 2 \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{4}\right).$$

Applying the identity again,

$$\sin\left(\frac{\pi}{4}\right) = 2 \sin\left(\frac{\pi}{8}\right) \cos\left(\frac{\pi}{8}\right),$$

so

$$1 = 2^2 \sin\left(\frac{\pi}{8}\right) \cos\left(\frac{\pi}{8}\right) \cos\left(\frac{\pi}{4}\right).$$

A1Continuing this process n times gives

$$1 = 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) \prod_{k=1}^n \cos\left(\frac{\pi}{2^{k+1}}\right).$$

A2From part **(c)(i)**,

$$P_n = \prod_{k=1}^n \cos\left(\frac{\pi}{2^{k+1}}\right).$$

Therefore,

$$1 = 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) P_n.$$

A1

$$\boxed{1 = 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) P_n}$$

[5 marks]

METHOD 2

Define the statement

$$S_n : \quad 1 = 2^n \sin \left(\frac{\pi}{2^{n+1}} \right) \prod_{k=1}^n \cos \left(\frac{\pi}{2^{k+1}} \right).$$

For $n = 1$,

$$2 \sin \left(\frac{\pi}{4} \right) \cos \left(\frac{\pi}{4} \right) = \sin \left(\frac{\pi}{2} \right) = 1.$$

A1

So S_1 is true.

Assume S_k is true for some integer $k \geq 1$, so

$$1 = 2^k \sin \left(\frac{\pi}{2^{k+1}} \right) \prod_{j=1}^k \cos \left(\frac{\pi}{2^{j+1}} \right).$$

M1

Now consider

$$2^{k+1} \sin \left(\frac{\pi}{2^{k+2}} \right) \prod_{j=1}^{k+1} \cos \left(\frac{\pi}{2^{j+1}} \right).$$

Separate the final factor in the product:

$$= 2^{k+1} \sin \left(\frac{\pi}{2^{k+2}} \right) \cos \left(\frac{\pi}{2^{k+2}} \right) \prod_{j=1}^k \cos \left(\frac{\pi}{2^{j+1}} \right).$$

M1

continued...

Using

$$2 \sin u \cos u = \sin(2u),$$

with

$$u = \frac{\pi}{2^{k+2}},$$

$$2 \sin \left(\frac{\pi}{2^{k+2}} \right) \cos \left(\frac{\pi}{2^{k+2}} \right) = \sin \left(\frac{\pi}{2^{k+1}} \right).$$

Therefore,

$$2^{k+1} \sin \left(\frac{\pi}{2^{k+2}} \right) \prod_{j=1}^{k+1} \cos \left(\frac{\pi}{2^{j+1}} \right) = 2^k \sin \left(\frac{\pi}{2^{k+1}} \right) \prod_{j=1}^k \cos \left(\frac{\pi}{2^{j+1}} \right).$$

A1

By the induction hypothesis, this is equal to 1.

Thus S_{k+1} is true.

Therefore, by induction,

$$1 = 2^n \sin \left(\frac{\pi}{2^{n+1}} \right) \prod_{k=1}^n \cos \left(\frac{\pi}{2^{k+1}} \right).$$

R1

Using part (c)(i),

$$1 = 2^n \sin \left(\frac{\pi}{2^{n+1}} \right) P_n$$

[5 marks]

METHOD 3

Using the double-angle identity,

$$\begin{aligned}\sin(2\theta) &= 2 \sin(\theta) \cos(\theta) \\ \therefore \cos(\theta) &= \frac{\sin(2\theta)}{2 \sin(\theta)}.\end{aligned}$$

A1

Substitute back into P_n :

$$P_n = \prod_{k=1}^n \frac{\sin\left(\frac{\pi}{2^k}\right)}{2 \sin\left(\frac{\pi}{2^{k+1}}\right)}.$$

M1

Expanding the product gives

$$\begin{aligned}P_n &= \frac{\sin\left(\frac{\pi}{2}\right)}{2 \sin\left(\frac{\pi}{4}\right)} \cdot \frac{\sin\left(\frac{\pi}{4}\right)}{2 \sin\left(\frac{\pi}{8}\right)} \cdot \frac{\sin\left(\frac{\pi}{8}\right)}{2 \sin\left(\frac{\pi}{16}\right)} \cdot \dots \cdot \frac{\sin\left(\frac{\pi}{2^n}\right)}{2 \sin\left(\frac{\pi}{2^{n+1}}\right)} \\ &= \frac{\sin\left(\frac{\pi}{2}\right)}{2 \cancel{\sin\left(\frac{\pi}{4}\right)}} \cdot \frac{\cancel{\sin\left(\frac{\pi}{4}\right)}}{2 \cancel{\sin\left(\frac{\pi}{8}\right)}} \cdot \frac{\cancel{\sin\left(\frac{\pi}{8}\right)}}{2 \cancel{\sin\left(\frac{\pi}{16}\right)}} \cdot \dots \cdot \frac{\cancel{\sin\left(\frac{\pi}{2^n}\right)}}{2 \sin\left(\frac{\pi}{2^{n+1}}\right)} \\ &= \frac{1}{2^n} \cdot \frac{\sin\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2^{n+1}}\right)} \\ &= \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)}.\end{aligned}$$

M1A1

Finally,

$$\begin{aligned}P_n &= \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)} \\ \therefore \boxed{1 &= 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) P_n}\end{aligned}$$

A1**[5 marks]**

Note: Accept products written in reverse order, since multiplication is commutative. Award follow-through marks from part **(c)(i)** where appropriate.

(iii) METHOD 1

From part **(c)(ii)**,

$$1 = 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) P_n.$$

Hence,

$$P_n = \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)}.$$

A1

Let

$$u_n = \frac{\pi}{2^{n+1}}.$$

Then $u_n \rightarrow 0$ as $n \rightarrow \infty$, and

$$2^n \sin\left(\frac{\pi}{2^{n+1}}\right) = 2^n \sin u_n.$$

Since

$$2^n = \frac{\pi}{2u_n},$$

we get

$$2^n \sin u_n = \frac{\pi}{2} \cdot \frac{\sin u_n}{u_n}.$$

M1

Using

$$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1,$$

$$\lim_{n \rightarrow \infty} 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) = \frac{\pi}{2}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P_n = \frac{1}{\pi/2} = \frac{2}{\pi}.$$

A1

$$\boxed{\lim_{n \rightarrow \infty} P_n = \frac{2}{\pi}}$$

[3 marks]

METHOD 2

From part (c)(ii),

$$P_n = \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)}.$$

A1

For large n ,

$$\frac{\pi}{2^{n+1}} \rightarrow 0$$

so

$$\sin\left(\frac{\pi}{2^{n+1}}\right) \approx \frac{\pi}{2^{n+1}}.$$

M1

Thus,

$$2^n \sin\left(\frac{\pi}{2^{n+1}}\right) \approx 2^n \cdot \frac{\pi}{2^{n+1}} = \frac{\pi}{2}.$$

Hence,

$$\lim_{n \rightarrow \infty} P_n = \frac{1}{\pi/2} = \frac{2}{\pi}.$$

A1

$\lim_{n \rightarrow \infty} P_n = \frac{2}{\pi}$

[3 marks]

Note: Accept equivalent limit arguments. Candidates must justify the limiting behaviour of the denominator.

(d) (i) METHOD 1

Using the given result,

$$b_k(x) = 2 \cos \left(\frac{x}{2^k} \right).$$

Therefore,

$$\frac{b_k(x)}{2} = \cos \left(\frac{x}{2^k} \right),$$

and so

$$Q_n(x) = \prod_{k=1}^n \cos \left(\frac{x}{2^k} \right).$$

A1

Now use

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

repeatedly:

$$\begin{aligned} \sin x &= 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) \\ &= 2^2 \sin \left(\frac{x}{2^2} \right) \cos \left(\frac{x}{2^2} \right) \cos \left(\frac{x}{2} \right) \end{aligned}$$

and continuing this process gives

$$\sin x = 2^n \sin \left(\frac{x}{2^n} \right) \prod_{k=1}^n \cos \left(\frac{x}{2^k} \right).$$

M1A1

Using the expression for $Q_n(x)$,

$$\sin x = 2^n \sin \left(\frac{x}{2^n} \right) Q_n(x).$$

Hence,

$$Q_n(x) = \frac{\sin x}{2^n \sin \left(\frac{x}{2^n} \right)}.$$

A1

$$Q_n(x) = \frac{\sin x}{2^n \sin \left(\frac{x}{2^n} \right)}$$

[4 marks]

METHOD 2

We prove the formula by induction.

For $n = 1$,

$$Q_1(x) = \frac{b_1(x)}{2} = \cos\left(\frac{x}{2}\right).$$

Also,

$$\frac{\sin x}{2 \sin\left(\frac{x}{2}\right)} = \frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} = \cos\left(\frac{x}{2}\right).$$

So the result is true for $n = 1$.

A1

Assume that, for some integer $n = k \geq 1$,

$$Q_k(x) = \frac{\sin x}{2^k \sin\left(\frac{x}{2^k}\right)}.$$

M1

Then

$$\begin{aligned} Q_{k+1}(x) &= Q_k(x) \cdot \frac{b_{k+1}(x)}{2} \\ &= \frac{\sin x}{2^k \sin\left(\frac{x}{2^k}\right)} \cos\left(\frac{x}{2^{k+1}}\right). \end{aligned}$$

M1

Using

$$\sin\left(\frac{x}{2^k}\right) = 2 \sin\left(\frac{x}{2^{k+1}}\right) \cos\left(\frac{x}{2^{k+1}}\right),$$

we get

$$Q_{k+1}(x) = \frac{\sin x}{2^k \cdot 2 \sin\left(\frac{x}{2^{k+1}}\right) \cos\left(\frac{x}{2^{k+1}}\right)} \cos\left(\frac{x}{2^{k+1}}\right).$$

Cancelling,

$$Q_{k+1}(x) = \frac{\sin x}{2^{k+1} \sin\left(\frac{x}{2^{k+1}}\right)}.$$

A1

Therefore, by induction,

$$Q_n(x) = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}$$

[4 marks]

METHOD 3

Using the given result,

$$b_k(x) = 2 \cos \left(\frac{x}{2^k} \right).$$

Therefore,

$$\frac{b_k(x)}{2} = \cos \left(\frac{x}{2^k} \right),$$

and so

$$Q_n(x) = \prod_{k=1}^n \cos \left(\frac{x}{2^k} \right).$$

A1

Now use

$$\begin{aligned} \sin(2\theta) &= 2 \sin \theta \cos \theta \\ \therefore \cos(\theta) &= \frac{\sin(2\theta)}{2 \sin(\theta)}. \end{aligned}$$

Substitute into $Q_n(x)$:

$$\begin{aligned} Q_n(x) &= \prod_{k=1}^n \cos \left(\frac{x}{2^k} \right) \\ &= \prod_{k=1}^n \frac{\sin \left(\frac{x}{2^{k-1}} \right)}{2 \sin \left(\frac{x}{2^k} \right)}. \end{aligned}$$

M1

Expanding the product gives

$$\begin{aligned} Q_n(x) &= \frac{\sin(x)}{2 \sin \left(\frac{x}{2} \right)} \cdot \frac{\sin \left(\frac{x}{2} \right)}{2 \sin \left(\frac{x}{4} \right)} \cdot \dots \cdot \frac{\sin \left(\frac{x}{2^{n-1}} \right)}{2 \sin \left(\frac{x}{2^n} \right)} \\ &= \frac{\sin(x)}{2 \cancel{\sin \left(\frac{x}{2} \right)}} \cdot \frac{\cancel{\sin \left(\frac{x}{2} \right)}}{2 \cancel{\sin \left(\frac{x}{4} \right)}} \cdot \dots \cdot \frac{\cancel{\sin \left(\frac{x}{2^{n-1}} \right)}}{2 \sin \left(\frac{x}{2^n} \right)} \\ &= \boxed{\frac{\sin(x)}{2^n \sin \left(\frac{x}{2^n} \right)}}. \end{aligned}$$

A2**[4 marks]**

Note: Accept any valid derivation using repeated double-angle identities. Follow-through should be awarded from the candidate's expression for $Q_n(x)$, provided the structure is mathematically consistent.

(ii) From the definition of $Q_n(x)$ and the given expression for $b_k(x)$,

$$Q_n(x) = \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right).$$

Therefore,

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \lim_{n \rightarrow \infty} Q_n(x).$$

M1

From part (d)(i),

$$Q_n(x) = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}.$$

Let

$$v_n = \frac{x}{2^n}.$$

Then $v_n \rightarrow 0$ as $n \rightarrow \infty$, and

$$2^n \sin\left(\frac{x}{2^n}\right) = 2^n \sin v_n = x \cdot \frac{\sin v_n}{v_n}.$$

M1

Using

$$\lim_{v \rightarrow 0} \frac{\sin v}{v} = 1,$$

$$\lim_{n \rightarrow \infty} 2^n \sin\left(\frac{x}{2^n}\right) = x.$$

Hence,

$$\lim_{n \rightarrow \infty} Q_n(x) = \frac{\sin x}{x}.$$

A1

Therefore,

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \frac{\sin x}{x}$$

[3 marks]

Note: Accept arguments using $\sin u \sim u$ as $u \rightarrow 0$. The condition $0 < x < \pi$ ensures the intermediate sine terms are non-zero but is **not** required for the marks.